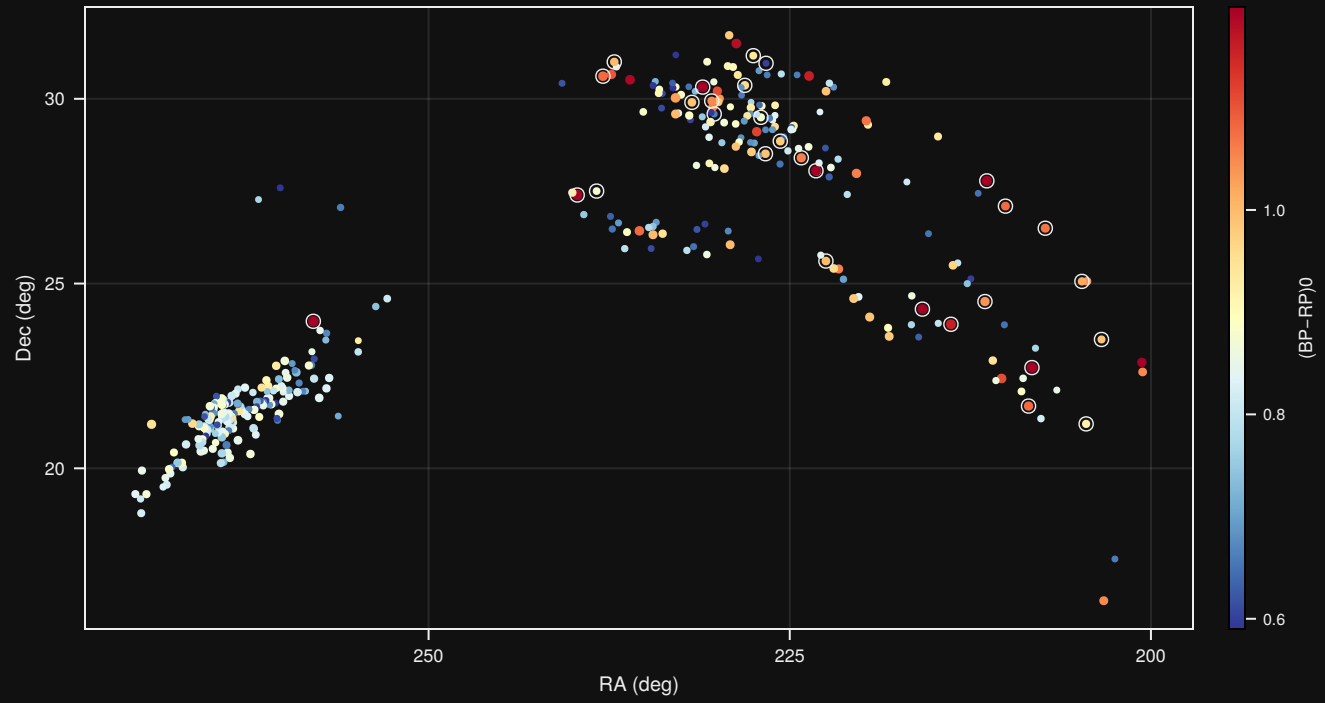
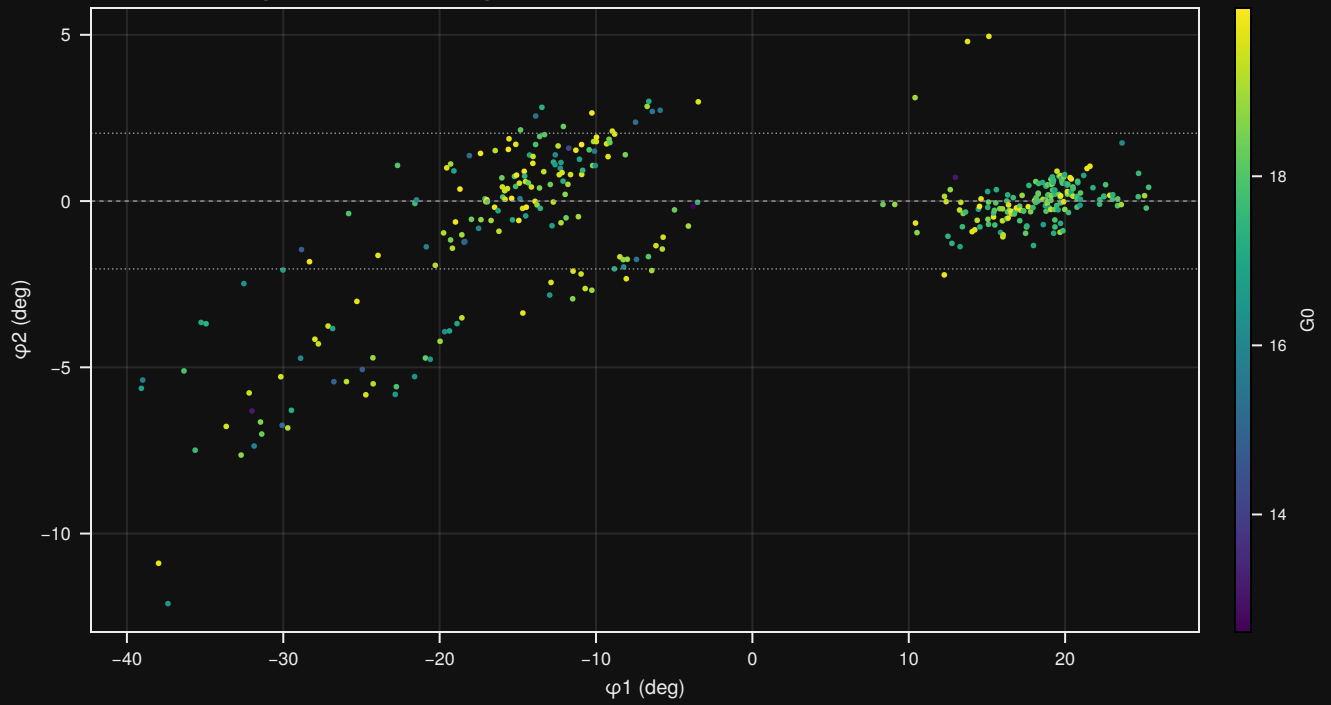
LMS-1

MEMBERSHIP	
Stars in catalog	358
with Gaia GSP-Phot	217 (61%)
with radial velocity	29
Stream id (sID)	51
GEOMETRY · great-circle fit	
Centroid RA, Dec	243.4, 23.5 deg
Centroid l, b	42.5, 50.6 deg
Pole RA, Dec	37.6, 60.6 deg
Length (φ1, 2–98%)	58.1 deg
Width σ(φ2), robust	1.02 deg
KINEMATICS	
pmRA* med ± σ	1.22 ± 1.76 mas/yr
pmDE med ± σ	-2.34 ± 1.52 mas/yr
μφ1* med ± σ	1.66 ± 2.41 mas/yr
μφ2 med ± σ	-1.91 ± 0.82 mas/yr
Parallax (cat.) med	0.102 ± 0.189 mas
Parallax (wtd, ZP-corr.)	0.098 ± 0.011 mas
Distance	d = 10.2 ± 1.1 kpc
Physical length	10.33 kpc
RV median ± σ(MAD)	-69.3 ± 67.5 km/s
RV gradient dV/dφ1	-4.96 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	12.6–20.0 (18.2)
(BP−RP)0 median	0.84
GSP-Phot [M/H] (n=217)	-1.35 ± 0.74
GSP-Phot Teff median	5329 K
RUWE > 1.4	2.3 %
PM clusters (largest)	14 (159 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	358 | 358
n_RV catalog | paper	29 | 29
<[Fe/H]> paper	-2.09
Stellar mass (paper)	15000 Msun
Pole offset vs paper	3.3 deg
Remark	Massive and wide system

Sky map · catalog members (n = 358) · size ∝ brightness · ring = has RV

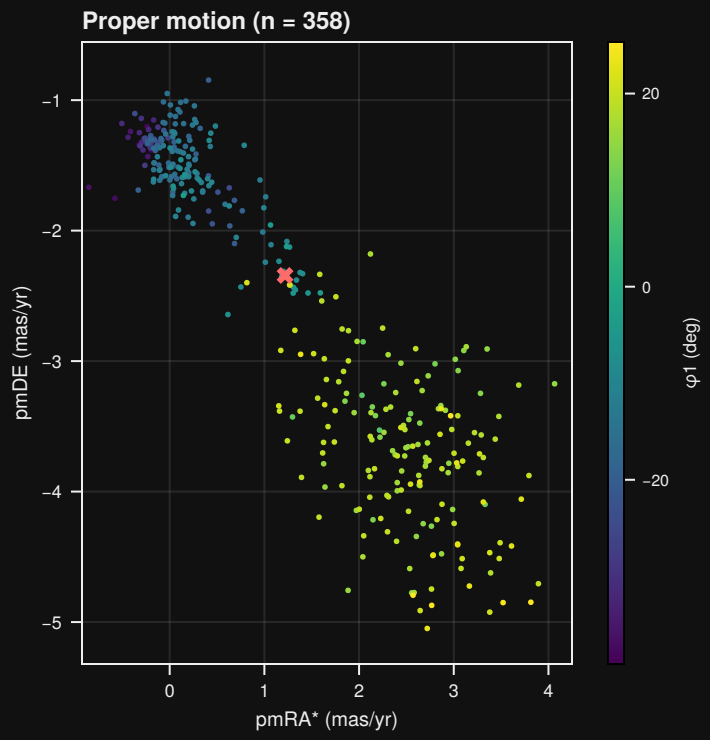


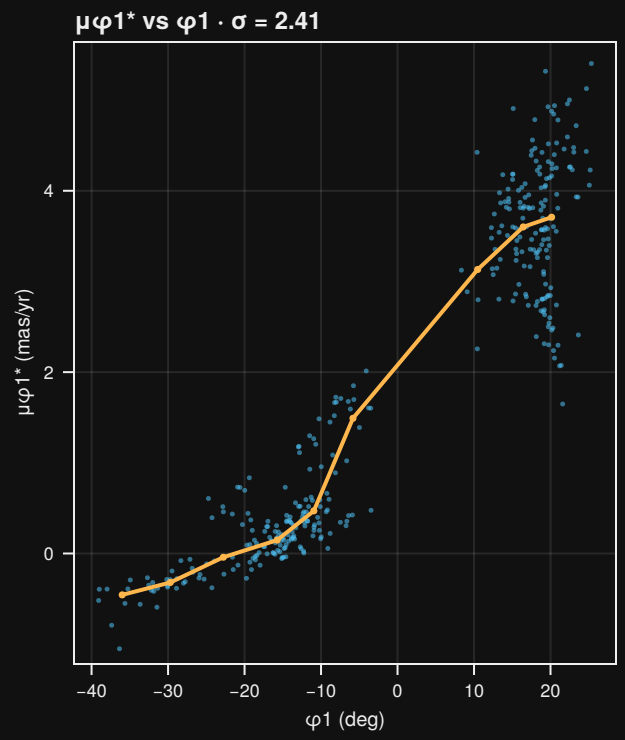
Stream frame (great-circle fit) · length 58.1° · width σ 1.02°

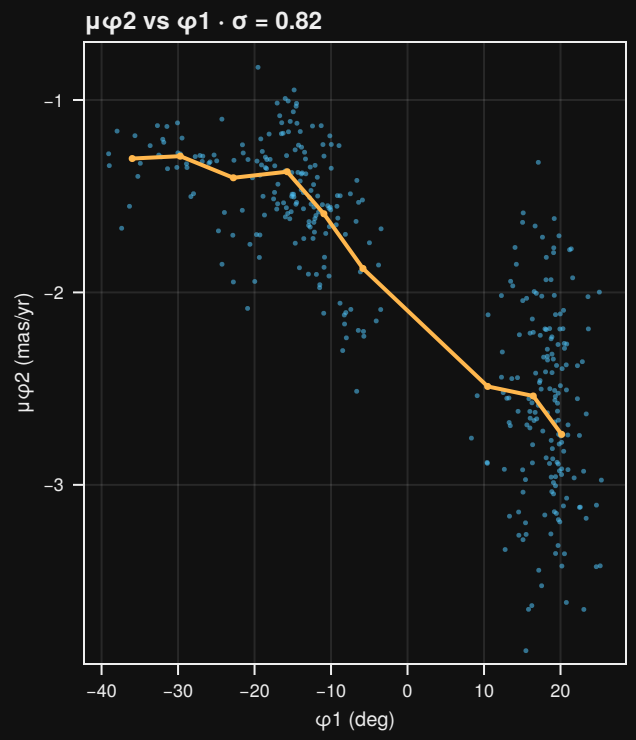


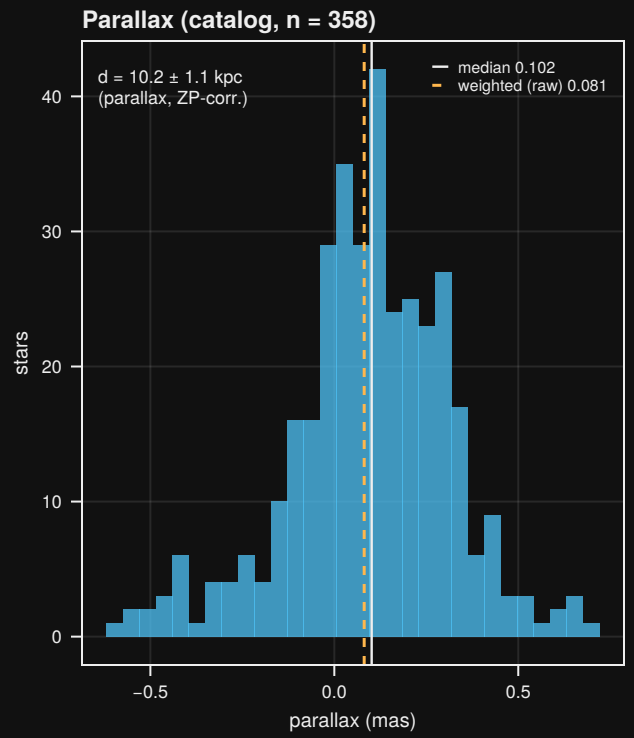
LMS-1

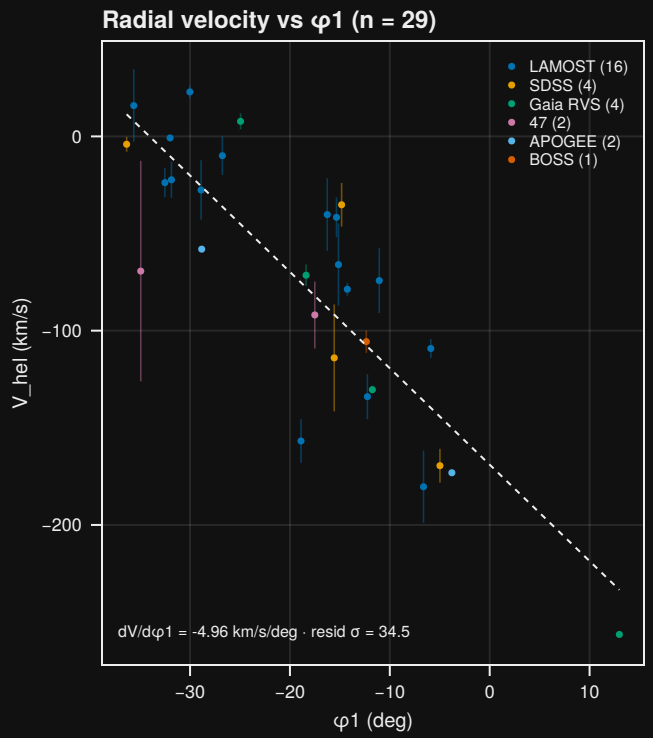
Kinematics · proper motion · parallax · radial velocity

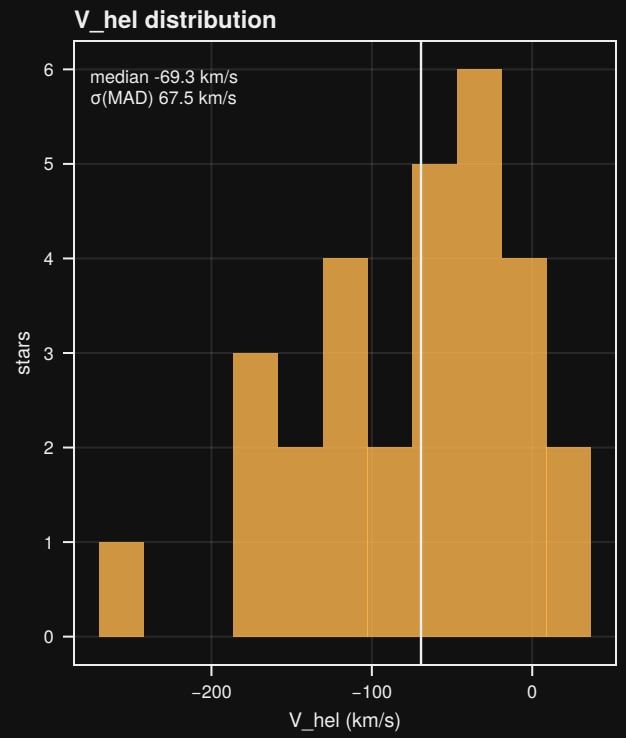






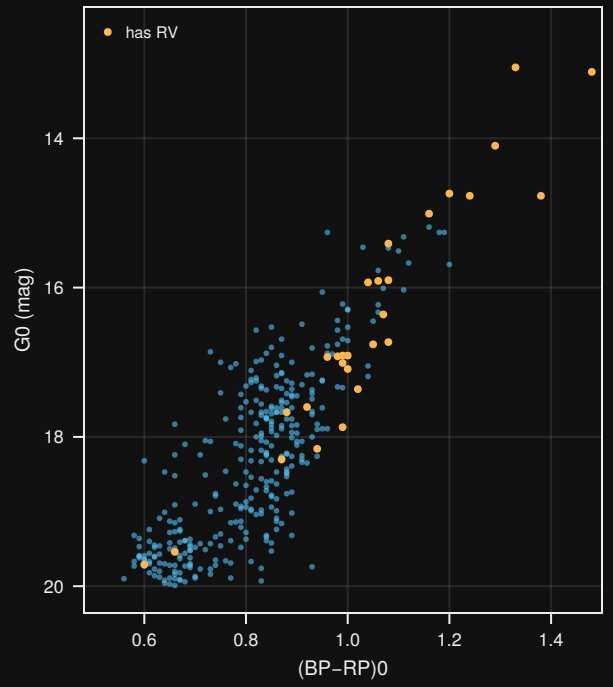




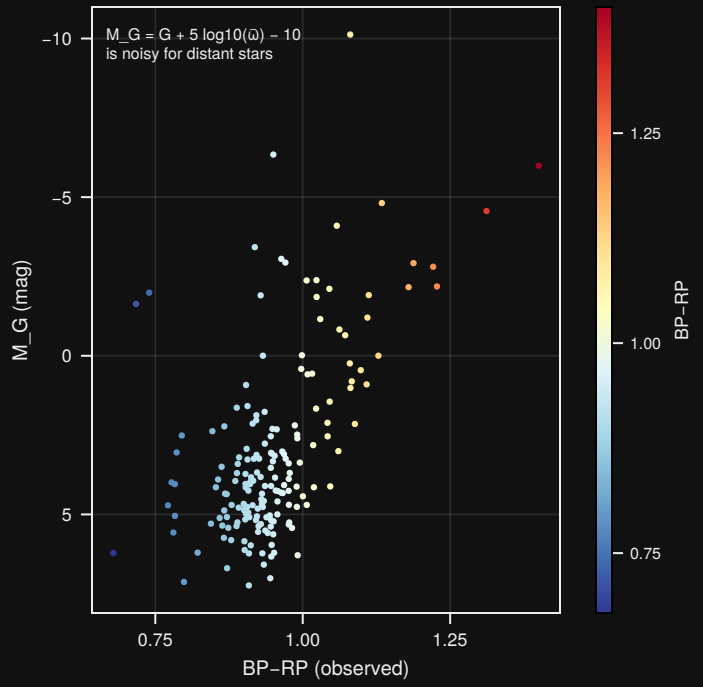


LMS-1

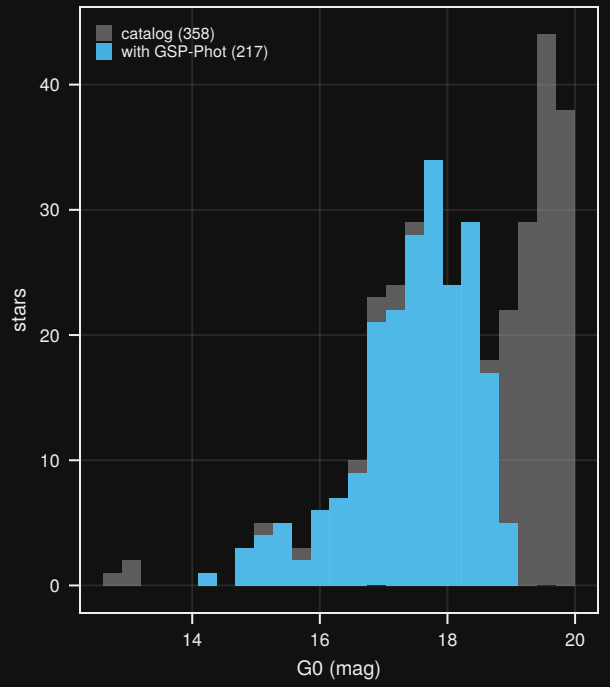
CMD · dereddened (n = 358)



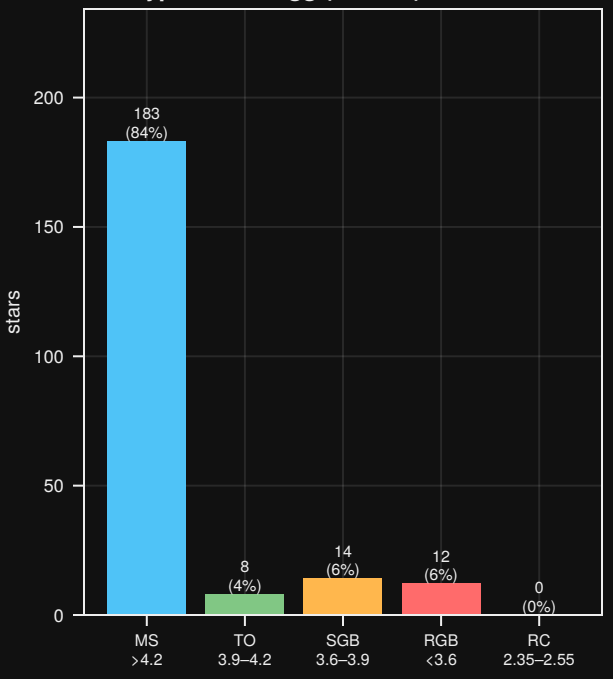
HR diagram · ϖ>0 (n = 180/217)



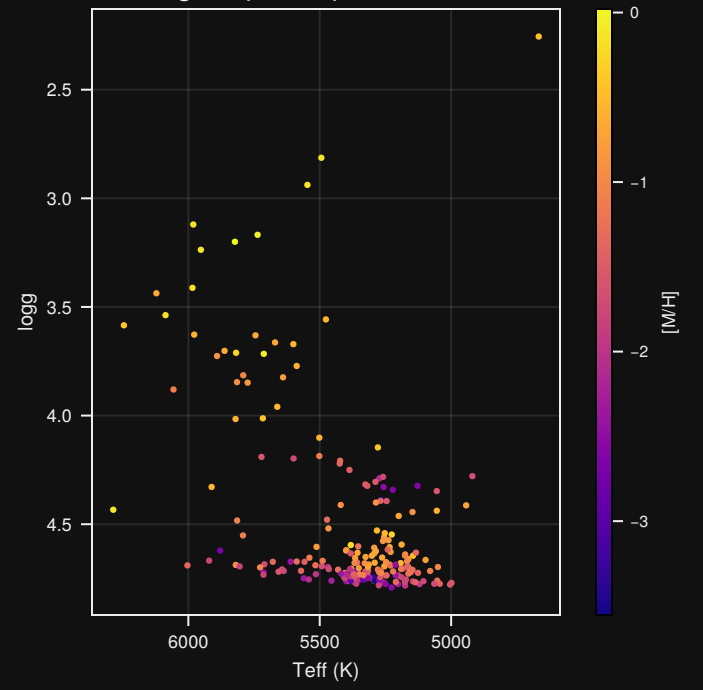
G0 & GSP-Phot coverage



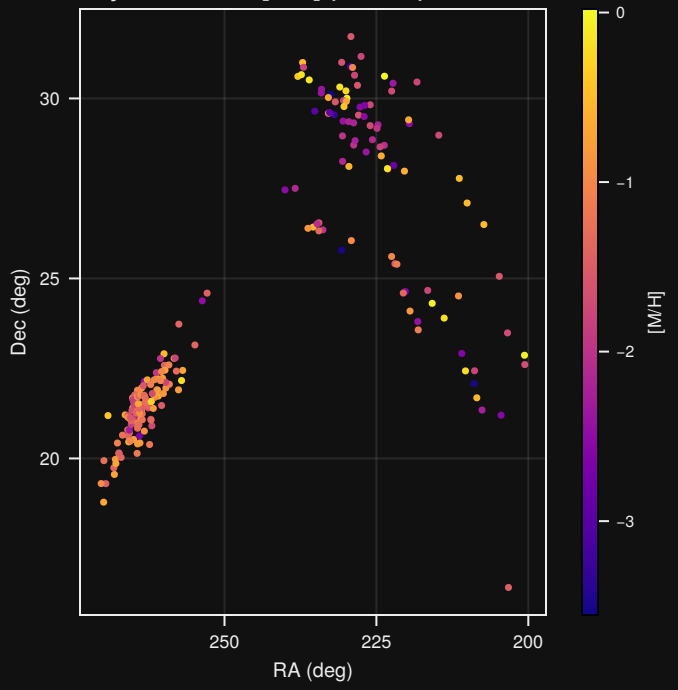
Star types from logg (n = 217)



Kiel diagram (n = 217)



Sky · GSP-Phot [M/H] (n = 217)

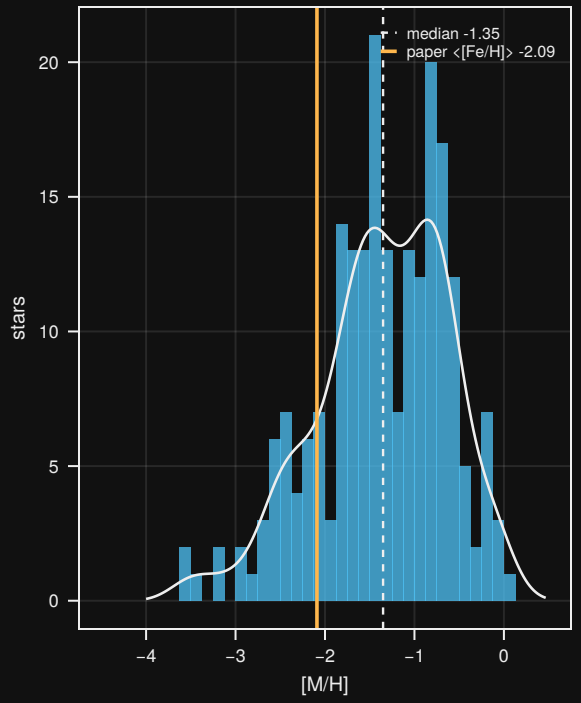


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

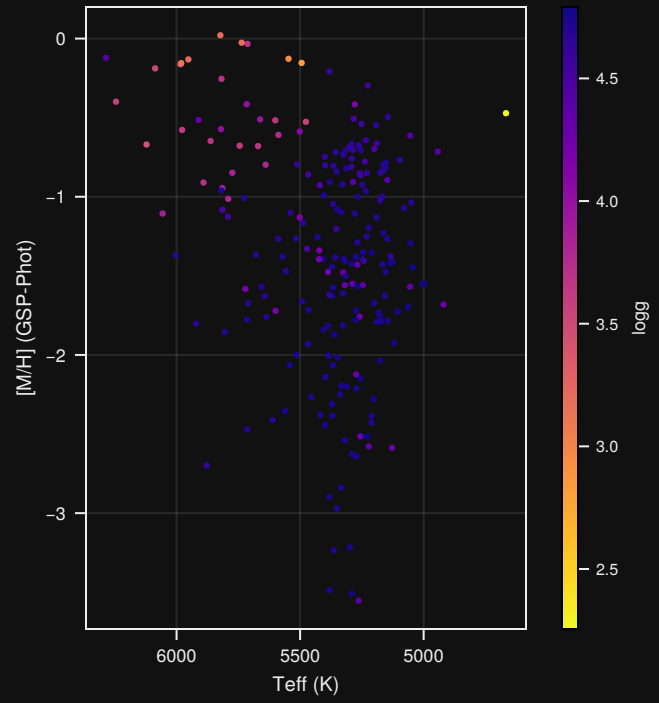
LMS-1

Metallicity · age · astrometric quality

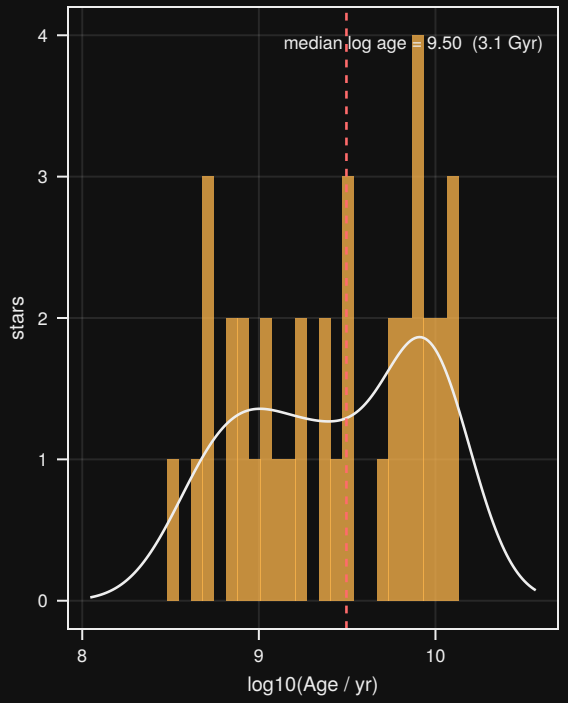
[M/H] GSP-Phot (n = 217)



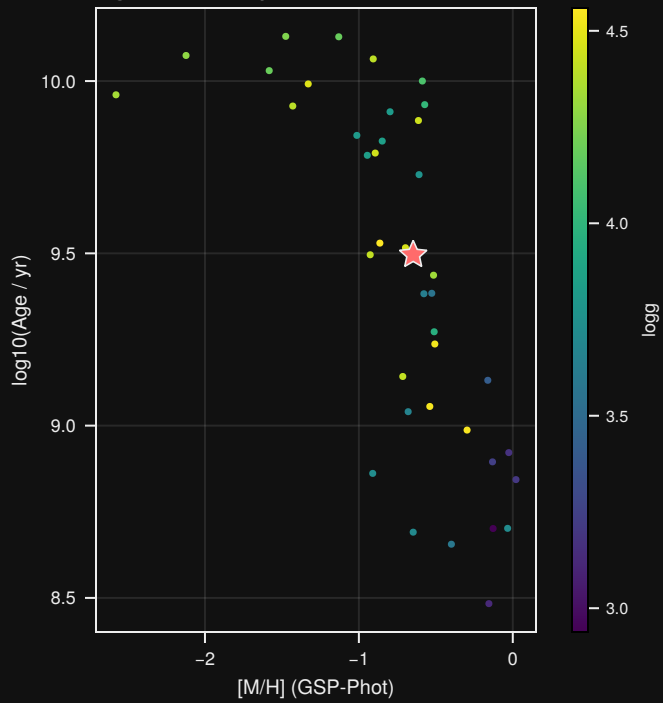
Teff vs [M/H] (n = 217)



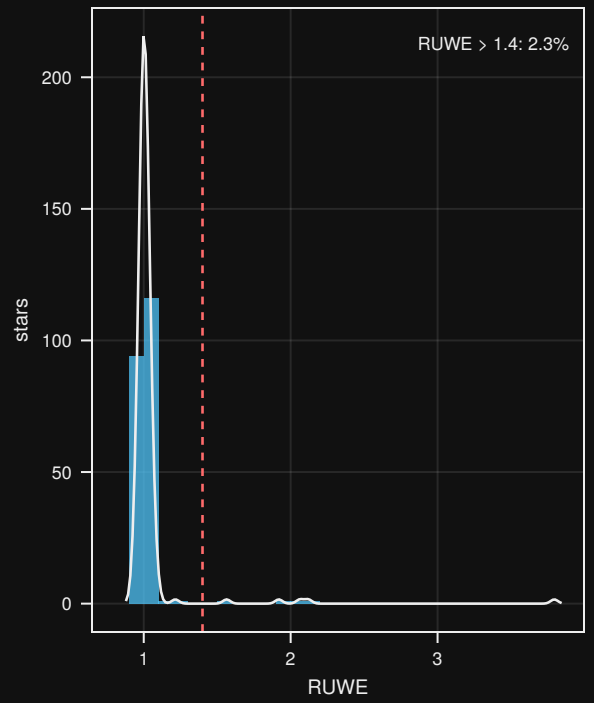
FLAME age (n = 39)



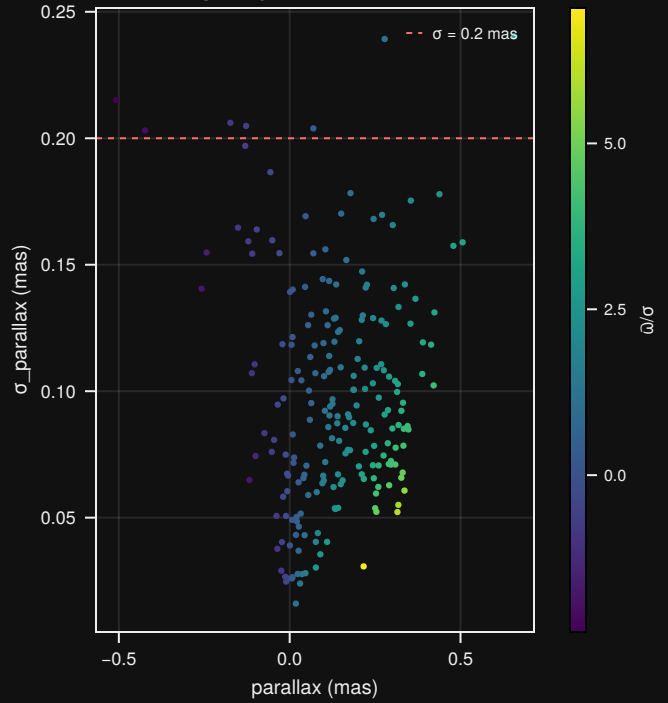
Age–metallicity (n = 39)



RUWE (n = 217)

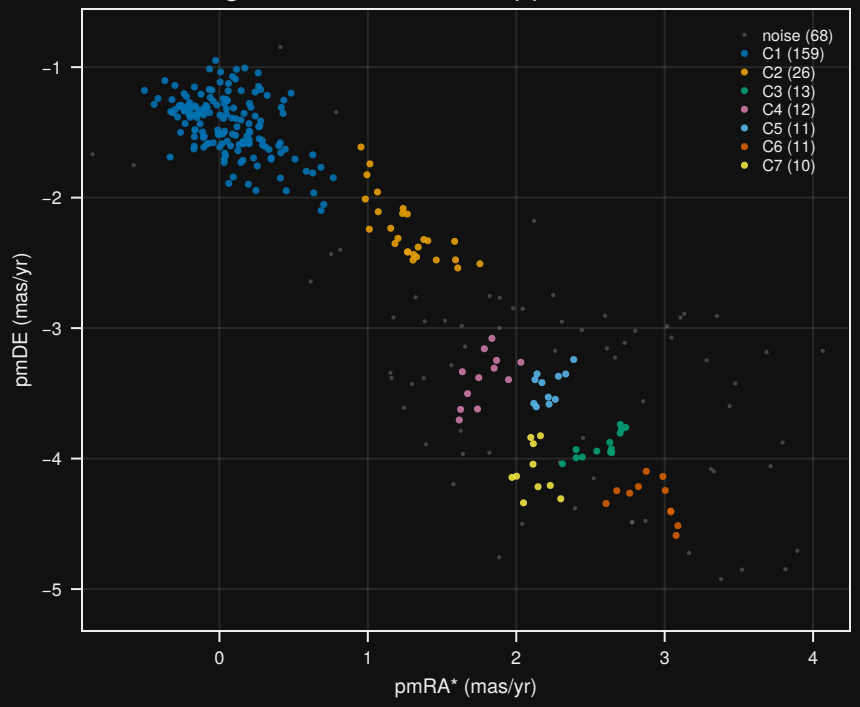


Parallax quality (n = 217)

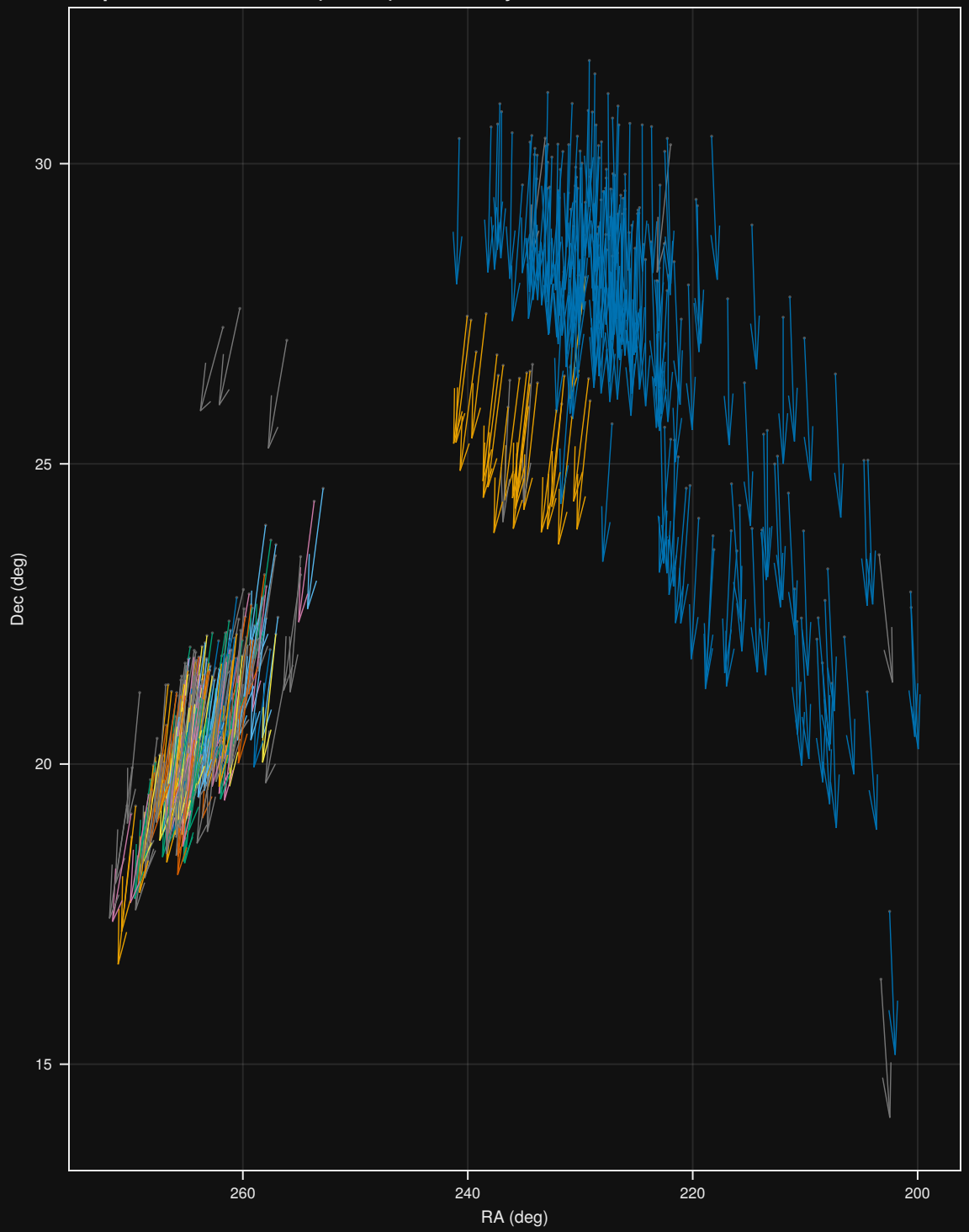


LMS-1

PM clustering · HDBSCAN · 14 cluster(s) · noise 19%



Proper-motion directions (arrows) coloured by cluster



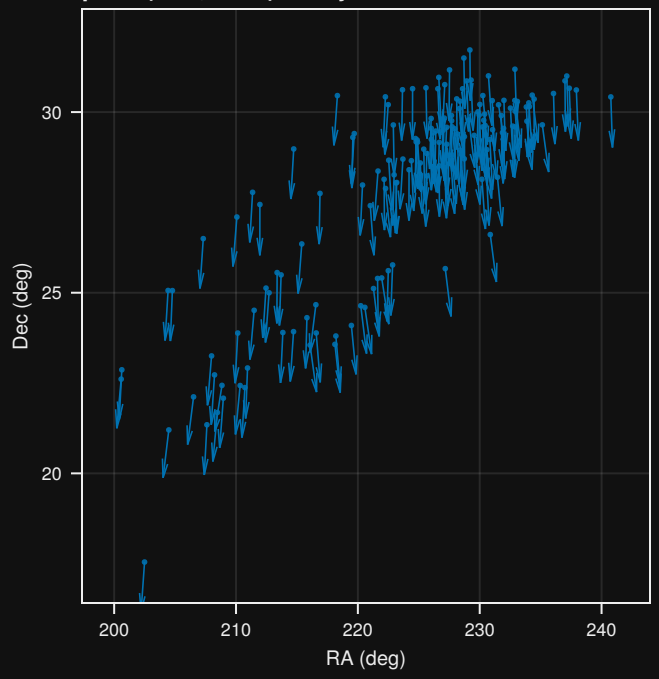
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	68	2.28	-3.21	1.13	-1.01	1
C1	159	0.04	-1.40	0.25	-1.73	25
C2	26	1.27	-2.33	0.19	-1.57	2
C3	13	2.63	-3.93	0.11	-1.26	0
C4	12	1.77	-3.36	0.14	-1.17	0
C5	11	2.22	-3.42	0.11	-1.35	1
C6	11	2.99	-4.26	0.15	-0.98	0
C7	10	2.11	-4.14	0.08	-1.27	0
C8	9	2.49	-3.66	0.13	-1.26	0

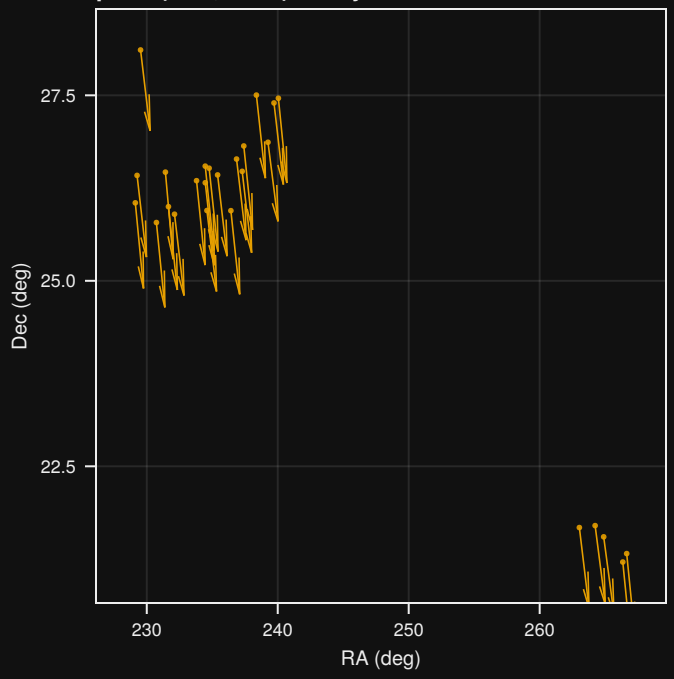
LMS-1

Per-cluster proper-motion vector maps

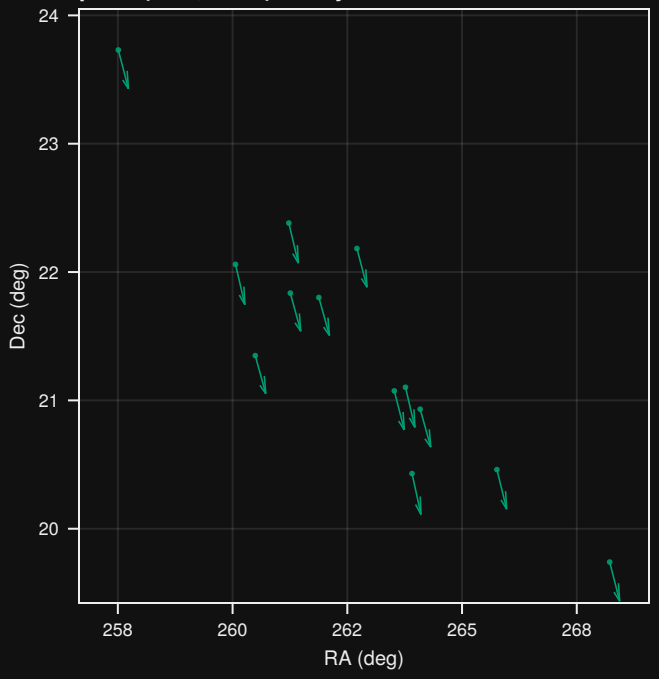
Cluster C1 · n = 159
pm = (0.04, -1.40) mas/yr



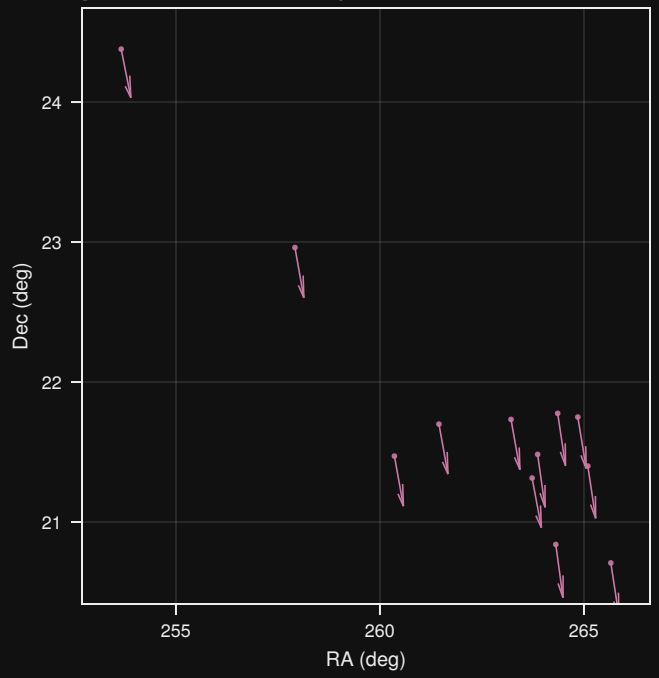
Cluster C2 · n = 26
pm = (1.27, -2.33) mas/yr



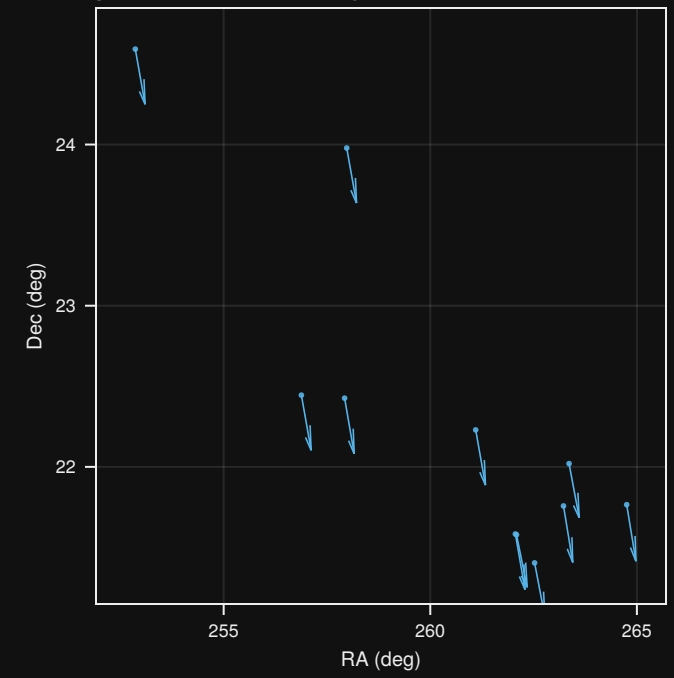
Cluster C3 · n = 13
pm = (2.63, -3.93) mas/yr



Cluster C4 · n = 12
pm = (1.77, -3.36) mas/yr



Cluster C5 · n = 11
pm = (2.22, -3.42) mas/yr



Noise · n = 68

